V. (points) Magnetic fields affect electrons, so the current-voltage relation may depart from the familiar Ohm's law. This problem studies one such example.

A 2D electron gas in plane has areal density (electrons per unit area). Along positive is a uniform electric field is the unit vector); along , the normal magnetic field is uniform, is the unit vector). The drag from scattering is proportional to velocity , written with time parameter as

is the electron mass. Find the steady and current densities (current per unit transverse length) and using charge magnitude and , where